stat_52700_project

Abir Mukherjee

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```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
insurance_orig <- read.csv("insurance.csv")
```

```r
glimpse(insurance_orig)
```

```
## Rows: 1,338
## Columns: 7
## $ age      <int> 19, 18, 28, 33, 32, 31, 46, 37, 37, 60, 25, 62, 23, 56, 27, 1~
## $ sex      <chr> "female", "male", "male", "male", "male", "female", "female",~
## $ bmi      <dbl> 27.900, 33.770, 33.000, 22.705, 28.880, 25.740, 33.440, 27.74~
## $ children <int> 0, 1, 3, 0, 0, 0, 1, 3, 2, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0~
## $ smoker   <chr> "yes", "no", "no", "no", "no", "no", "no", "no", "no", "no", ~
## $ region   <chr> "southwest", "southeast", "southeast", "northwest", "northwes~
## $ charges  <dbl> 16884.924, 1725.552, 4449.462, 21984.471, 3866.855, 3756.622,~
```

```r
insurance <- as_tibble(insurance_orig)
glimpse(insurance)
```

```
## Rows: 1,338
## Columns: 7
## $ age      <int> 19, 18, 28, 33, 32, 31, 46, 37, 37, 60, 25, 62, 23, 56, 27, 1~
## $ sex      <chr> "female", "male", "male", "male", "male", "female", "female",~
## $ bmi      <dbl> 27.900, 33.770, 33.000, 22.705, 28.880, 25.740, 33.440, 27.74~
## $ children <int> 0, 1, 3, 0, 0, 0, 1, 3, 2, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0~
## $ smoker   <chr> "yes", "no", "no", "no", "no", "no", "no", "no", "no", "no", ~
## $ region   <chr> "southwest", "southeast", "southeast", "northwest", "northwes~
## $ charges  <dbl> 16884.924, 1725.552, 4449.462, 21984.471, 3866.855, 3756.622,~
```

```r
colnames(insurance)
```

```
## [1] "age"      "sex"      "bmi"      "children" "smoker"   "region"   "charges"
```

```r
dim(insurance)
```

```
## [1] 1338    7
```

```r
colSums(is.na(insurance))
```

```
##      age      sex      bmi children   smoker   region  charges
##        0        0        0        0        0        0        0
```

```r
insurance %>%
  duplicated() %>%
  sum()
```

```
## [1] 1
```

```r
insurance_clean <- insurance %>%
  distinct()
insurance_clean <- insurance_clean %>%
  mutate(
    sex = as.factor(sex),
    smoker = as.factor(smoker),
    region = as.factor(region)
  )

glimpse(insurance_clean)
```

```
## Rows: 1,337
## Columns: 7
## $ age      <int> 19, 18, 28, 33, 32, 31, 46, 37, 37, 60, 25, 62, 23, 56, 27, 1~
## $ sex      <fct> female, male, male, male, male, female, female, female, male,~
## $ bmi      <dbl> 27.900, 33.770, 33.000, 22.705, 28.880, 25.740, 33.440, 27.74~
## $ children <int> 0, 1, 3, 0, 0, 0, 1, 3, 2, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0~
## $ smoker   <fct> yes, no, no, no, no, no, no, no, no, no, no, yes, no, no, yes~
## $ region   <fct> southwest, southeast, southeast, northwest, northwest, southe~
## $ charges  <dbl> 16884.924, 1725.552, 4449.462, 21984.471, 3866.855, 3756.622,~
```

```r
insurance_clean <- insurance_clean %>%
  mutate(
    obese = bmi >= 30,
    age_group = case_when(
      age < 30 ~ "young",
      age < 50 ~ "middle",
      TRUE ~ "older"
    )
  )
head(insurance_clean)
```

```
## # A tibble: 6 x 9
##     age sex      bmi children smoker region    charges obese age_group
##   <int> <fct>  <dbl>    <int> <fct>  <fct>       <dbl> <lgl> <chr>
## 1    19 female  27.9        0 yes    southwest  16885. FALSE young
## 2    18 male    33.8        1 no     southeast   1726. TRUE  young
## 3    28 male    33          3 no     southeast   4449. TRUE  young
## 4    33 male    22.7        0 no     northwest  21984. FALSE middle
## 5    32 male    28.9        0 no     northwest   3867. FALSE middle
## 6    31 female  25.7        0 no     southeast   3757. FALSE middle
```

```r
insurance_clean %>%
  summarise(
    count = n(),
    mean_age = mean(age),
    mean_bmi = mean(bmi),
    mean_children = mean(children),
    mean_charges = mean(charges),
    median_charges = median(charges),
    sd_charges = sd(charges)
  )
```

```
## # A tibble: 1 x 7
##   count mean_age mean_bmi mean_children mean_charges median_charges sd_charges
##   <int>    <dbl>    <dbl>         <dbl>        <dbl>          <dbl>      <dbl>
## 1  1337     39.2     30.7          1.10       13279.          9386.     12110.
```

```r
# Average charges by smoking status
insurance_clean %>%
  group_by(smoker) %>%
  summarise(
    count = n(),
    mean_charges = mean(charges),
    median_charges = median(charges),
    .groups = "drop"
  )
```

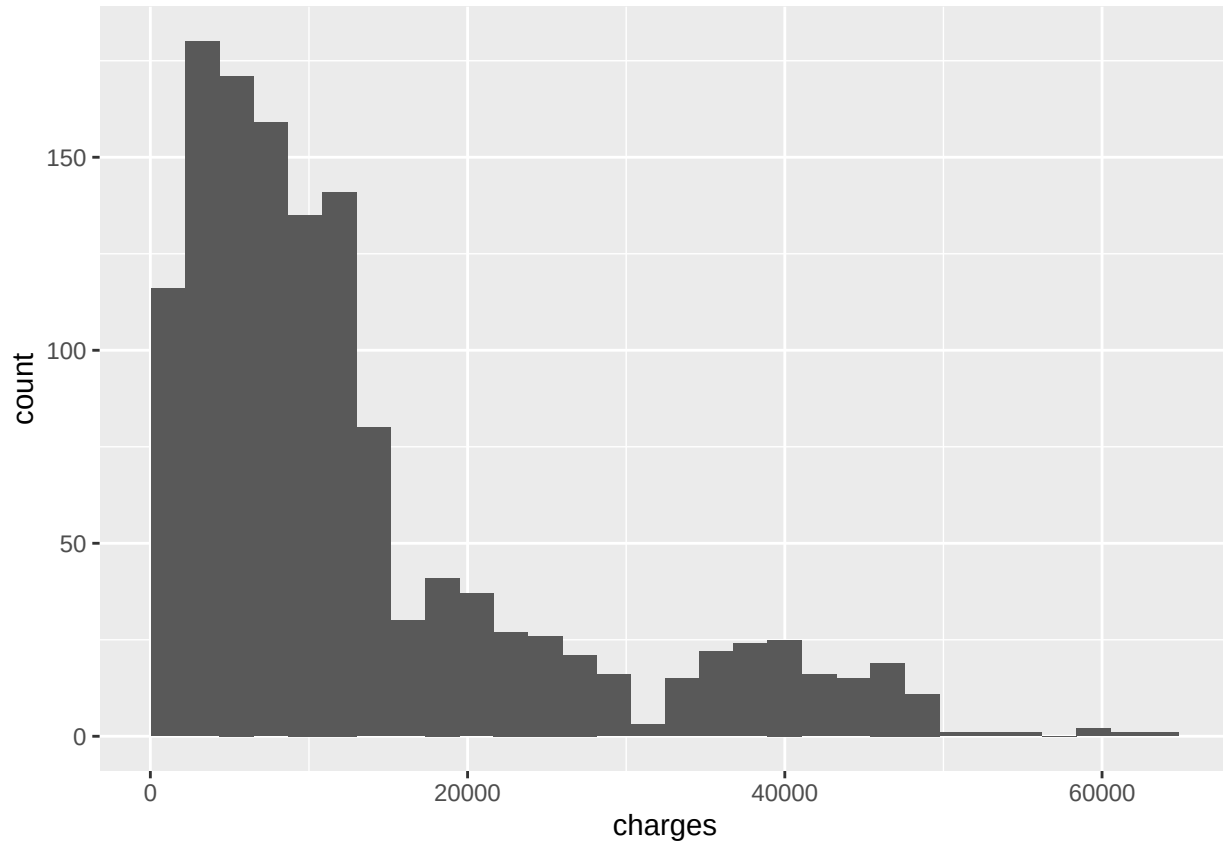
```
## # A tibble: 2 x 4
##   smoker count mean_charges median_charges
##   <fct>  <int>        <dbl>          <dbl>
## 1 no      1063        8441.          7346.
## 2 yes      274       32050.         34456.
```

```r
# Average charges by age group and smoker
insurance_clean %>%
  group_by(age_group, smoker) %>%
  summarise(
    count = n(),
    mean_charges = mean(charges),
    median_charges = median(charges),
    .groups = "drop"
  )
```

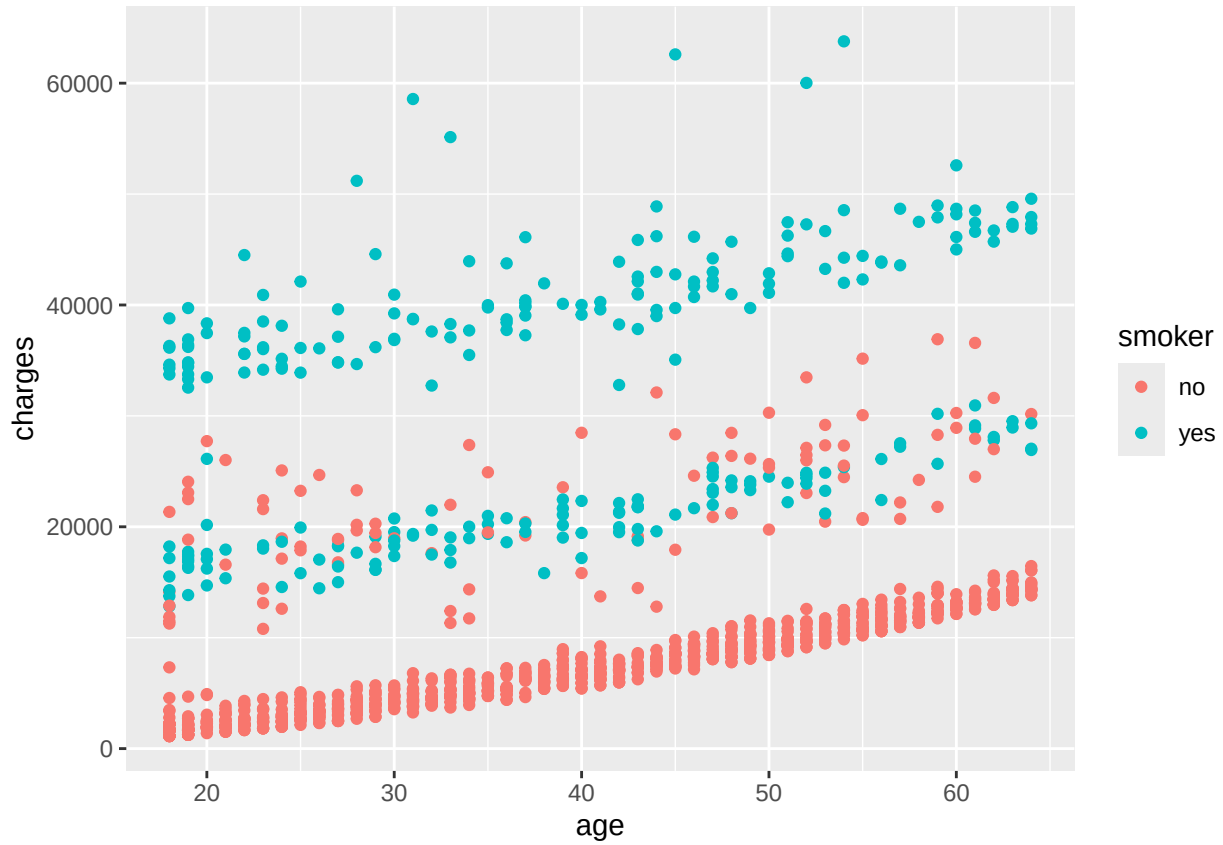
```
## # A tibble: 6 x 5
```

```
##   age_group smoker count mean_charges median_charges
##   <chr>     <fct>  <int>        <dbl>          <dbl>
## 1 middle    no       416        7822.          7046.
## 2 middle    yes      120       31503.         35281.
## 3 older     no       317       13431.         12029.
## 4 older     yes       68       38748.         43696.
## 5 young     no       330        4427.          2709.
## 6 young     yes       86       27518.         33604.
```

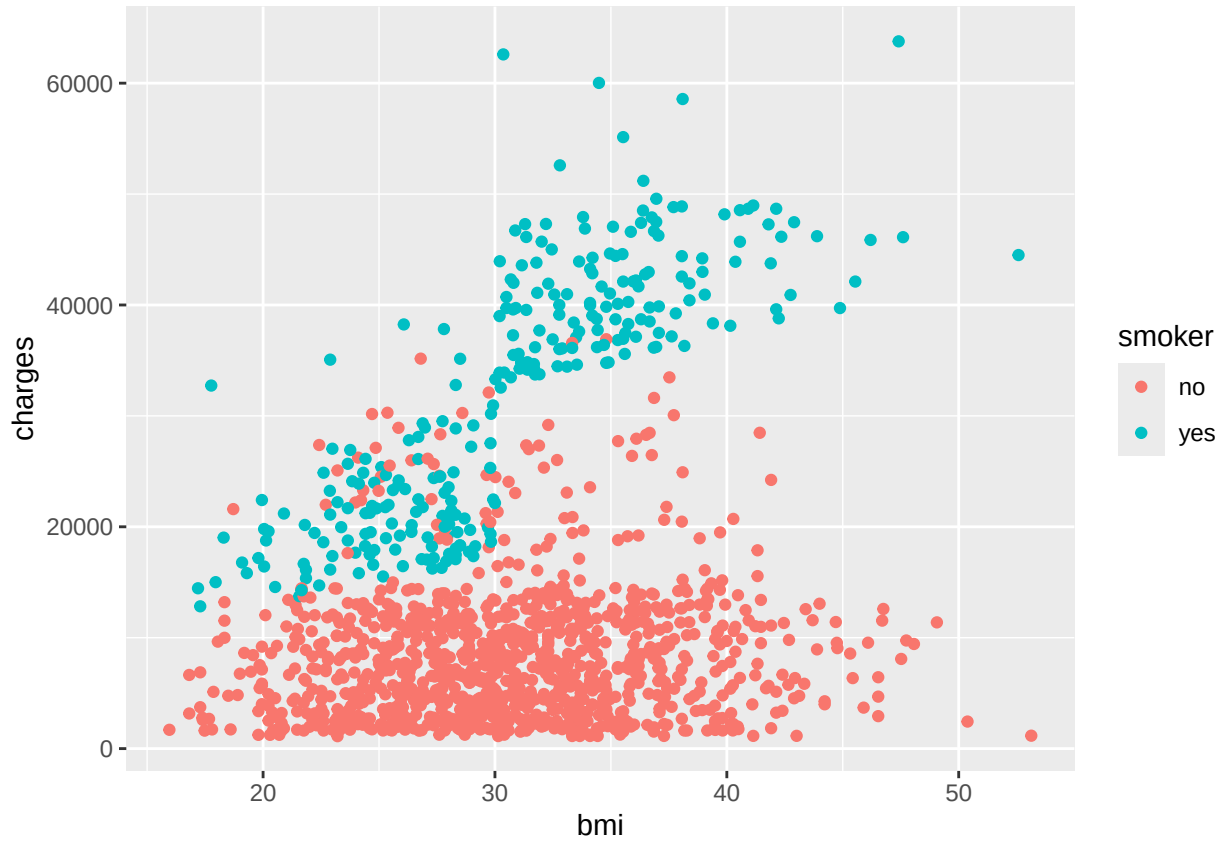
```
ggplot(insurance_clean, aes(x = charges)) +
  geom_histogram(bins = 30)
```



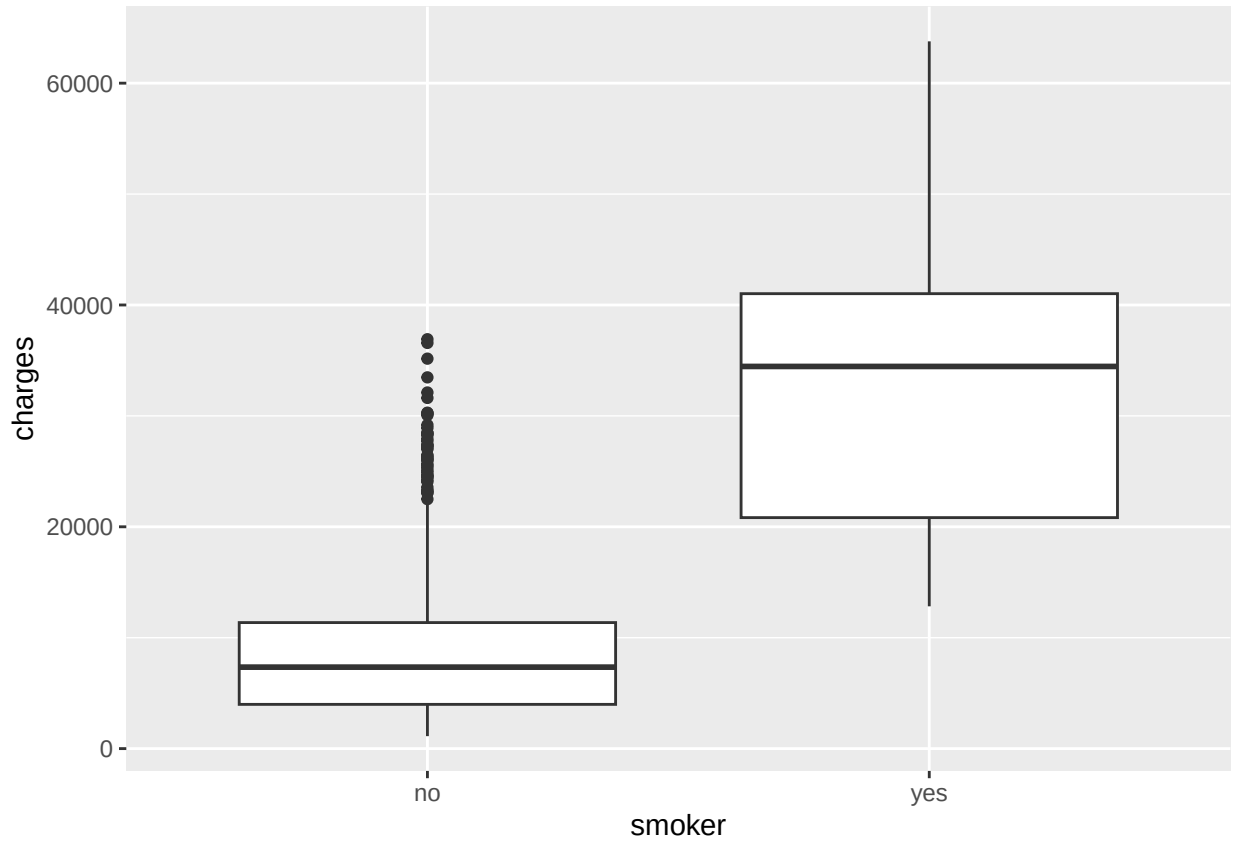

```
# Scatterplot: age vs charges
ggplot(insurance_clean, aes(x = age, y = charges, color=smoker)) +
  geom_point()
```



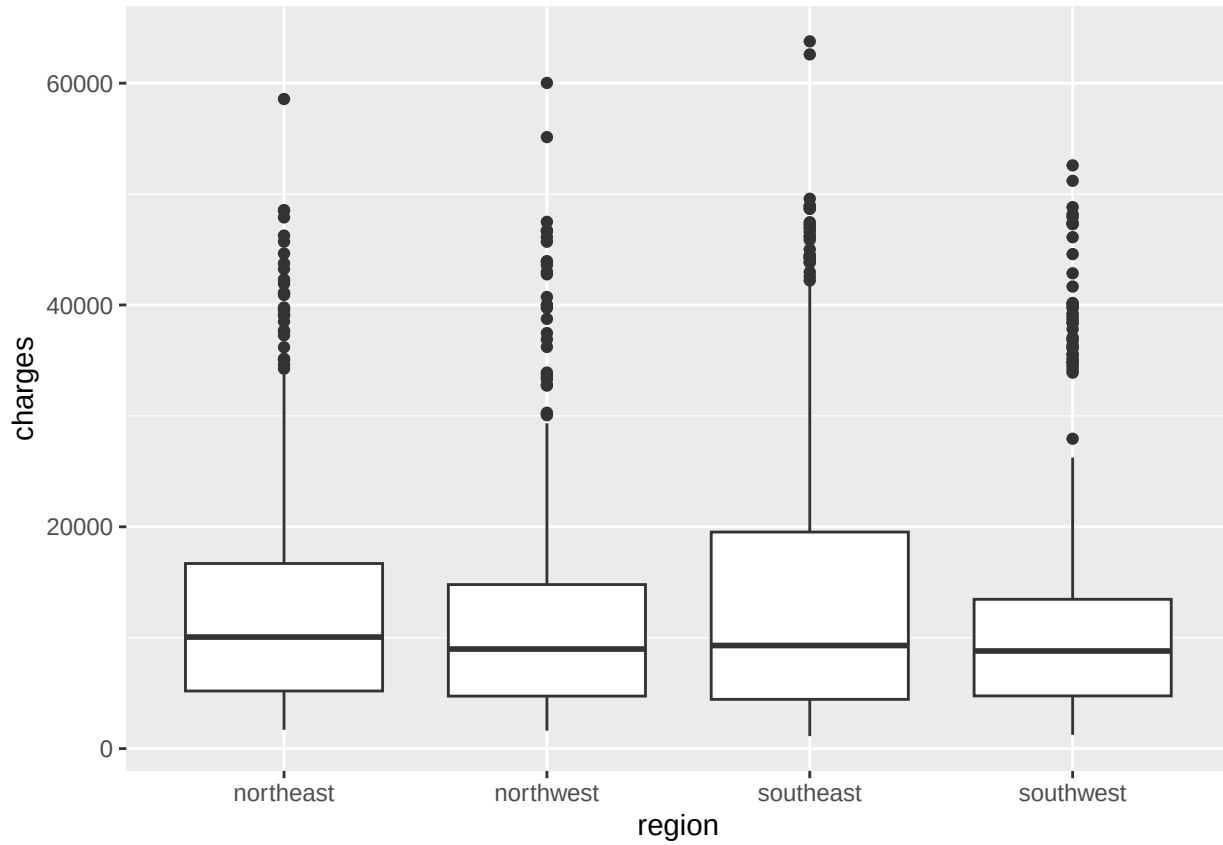

```r
# Scatterplot: bmi vs charges
ggplot(insurance_clean, aes(x = bmi, y = charges, color=smoker)) +
  geom_point()
```




```r
# Boxplot: charges by smoker
ggplot(insurance_clean, aes(x = smoker, y = charges)) +
  geom_boxplot()
```




```r
# Boxplot: charges by region
ggplot(insurance_clean, aes(x = region, y = charges)) +
  geom_boxplot()
```




```
numeric_data <- insurance_clean %>%
  select(age, bmi, children, charges)

cor_matrix <- cor(numeric_data)

round(cor_matrix, 2)
```

```
##           age  bmi children charges
## age      1.00 0.11     0.04    0.30
## bmi      0.11 1.00     0.01    0.20
## children 0.04 0.01     1.00    0.07
## charges  0.30 0.20     0.07    1.00
```

Linear regression. One predictor (age)

```
fit_age <- lm(charges ~ age, data = insurance_clean)

fit_age
```

```
##
## Call:
## lm(formula = charges ~ age, data = insurance_clean)
```

```
##
## Coefficients:
## (Intercept)          age
##      3190.0        257.2
```

```r
summary(fit_age)
```

```
##
## Call:
## lm(formula = charges ~ age, data = insurance_clean)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -8064  -6684  -5943   5466  47828
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3190.02     938.40   3.399 0.000695 ***
## age           257.23      22.53  11.419  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11560 on 1335 degrees of freedom
## Multiple R-squared:  0.08899,    Adjusted R-squared:  0.08831
## F-statistic: 130.4 on 1 and 1335 DF,  p-value: < 2.2e-16
```

Multiple regression

```r
fit_lm <- lm(charges ~ age + bmi + children + sex + smoker + region,
             data = insurance_clean)

fit_lm
```

```
##
## Call:
## lm(formula = charges ~ age + bmi + children + sex + smoker +
##     region, data = insurance_clean)
##
## Coefficients:
##     (Intercept)              age              bmi         children
##        -11936.6            256.8            339.3            474.8
##         sexmale        smokeryes  regionnorthwest  regionsoutheast
##          -129.5          23847.3           -349.2          -1035.3
## regionsouthwest
##          -960.1
```

```r
summary(fit_lm)
```

```
##
```

```
## Call:
## lm(formula = charges ~ age + bmi + children + sex + smoker +
##     region, data = insurance_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11305.1  -2850.3   -979.9   1395.0  29992.8
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)     -11936.56     988.23 -12.079  < 2e-16 ***
## age                256.76      11.91  21.555  < 2e-16 ***
## bmi                339.25      28.61  11.857  < 2e-16 ***
## children           474.82     137.90   3.443 0.000593 ***
## sexmale           -129.48     333.20  -0.389 0.697630
## smokeryes        23847.33     413.35  57.693  < 2e-16 ***
## regionnorthwest   -349.23     476.82  -0.732 0.464053
## regionsoutheast  -1035.27     478.87  -2.162 0.030804 *
## regionsouthwest   -960.08     478.11  -2.008 0.044836 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6064 on 1328 degrees of freedom
## Multiple R-squared:  0.7507, Adjusted R-squared:  0.7492
## F-statistic:   500 on 8 and 1328 DF,  p-value: < 2.2e-16
```

Splitting data to Test/Train

```r
set.seed(1)

train_ids <- sample(
  1:nrow(insurance_clean),
  size = round(0.70 * nrow(insurance_clean))
)

train <- insurance_clean[train_ids, ]
test <- insurance_clean[-train_ids, ]
```

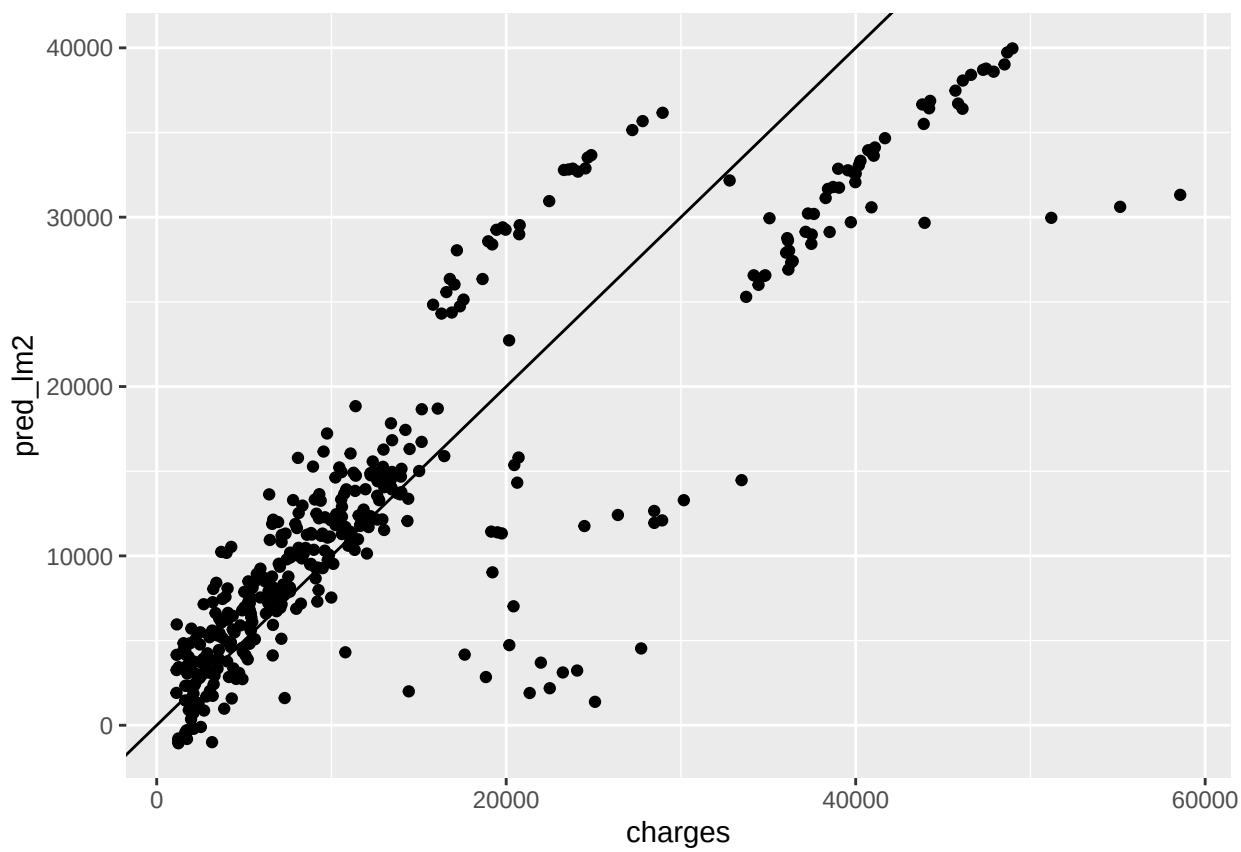
Comparing two models with numeric only and numeric + smoker

```r
fit_lm1 <- lm(charges ~ age + bmi + children, data = train)
fit_lm2 <- lm(charges ~ age + bmi + children + smoker, data = train)
test <- test %>%
  mutate(
    pred_lm1 = predict(fit_lm1, newdata = test),
    pred_lm2 = predict(fit_lm2, newdata = test)
  )
```

```r
mse <- function(actual, predicted) {
  mean((actual - predicted)^2)
}

model_results <- tibble(
  model = c("Linear regression: numeric predictors only",
            "Linear regression: numeric + smoker"),
  test_mse = c(
    mse(test$charges, test$pred_lm1),
    mse(test$charges, test$pred_lm2)
  )
)

model_results
```

```
## # A tibble: 2 x 2
##   model                                          test_mse
##   <chr>                                             <dbl>
## 1 Linear regression: numeric predictors only 151402615.
## 2 Linear regression: numeric + smoker         39081888.
```

```r
ggplot(test, aes(x = charges, y = pred_lm2)) +
  geom_point() +
  geom_abline()
```